Fill Rules, Normalization, and Chance Floors in Subject-Grouped ERP Decoding: A Five-Dataset Study with a Spiking Reservoir Reference

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Abstract*—Comparisons of EEG representations under channel loss are usually reported as one number per encoder, but that number depends on choices that are rarely stated: how a missing channel is filled, how each epoch is normalized, and how accuracy retention is scored. We study these choices on five event-related potential (ERP) datasets: two affective picture-viewing cohorts (211 and 228 participants) and three ERP CORE paradigms (P3, N400, N170). Encoders range from conventional late positive potential (LPP) and window amplitudes to band power, a matched six-bin amplitude code, and a fixed leaky integrate-and-fire reservoir read out through six spike-count bins. We prove that zero fill is not invariant to a change of feature origin and can drive a non-centered encoder to chance while mean fill is unaffected, and that per-epoch normalization of averaged ERPs rescales conditions by their trial counts. Both effects are measured: whole-epoch z-scoring let post-stimulus signal reach a pre-stimulus reservoir window that carries no affective information on its own (balanced accuracy 0.487 versus 0.305 under baseline-only scaling), and baseline z-scoring removed the oddball P3 effect from a conventional LPP score (paired d_z 1.05 to 0.05). With its single input gain chosen by nested cross-validation, the training-free reservoir reached 0.889 balanced accuracy on the oddball P3 task, above EEGNet (0.847) and a Riemannian tangent-space classifier (0.875) in point estimate, and exceeded the conventional LPP score by $+0.181$ [$+0.083$, $+0.278$]. [PENDING: gain-tuned results on N400, N170 and the two affective cohorts, and paired contrasts against the trained models]. We give a reporting checklist and release code that reproduces the conference results exactly.

***Index Terms*—EEG, event-related potentials, channel loss, imputation, normalization, spiking reservoir, evaluation protocol.**

I. Introduction

ELECTRODE loss is common in practical EEG, and a growing literature compares how well learned and fixed representations tolerate it [2]–[4]. Such comparisons rest on a scoring protocol: a rule for what a missing channel contributes, a normalization applied before encoding, and a summary of how much accuracy survives. Small changes to that protocol can reorder encoders. A recent foundation-model benchmark found that apparent channel-dropout fragility depends on whether dropped channels are removed or zero-padded [4], and preprocessing choices more broadly change deep-learning EEG results [5].

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This paper extends the authors' IEEE BIBM 2026 Doctoral Forum paper [1]. New material is listed in Section VI.

Our conference paper showed two such effects on one affective ERP cohort [1]. Raw retention has a chance floor that flatters weak encoders, and zero fill of a dropped channel is not invariant to an information-preserving re-centering of the features. A fixed spiking reservoir served there as a near-centered reference, because its embedding barely moves under constant fills.

This paper asks whether those conclusions hold beyond one cohort, one paradigm and one preprocessing pipeline, and what else in the protocol can decide a comparison. The contributions are:

1) **Theory.** Proposition 3 shows that a change of feature origin, which carries no information, moves zero-fill accuracy anywhere between its mean-fill value and chance. Proposition 4 shows that per-epoch normalization of averaged ERPs rescales each condition by the square root of its trial count, which can cancel a real effect (Section III).
2) **Two normalization artifacts, measured.** Whole-epoch z-scoring carried post-stimulus information into the mostly pre-stimulus window used by the conference reservoir, and baseline z-scoring cancelled the oddball P3 effect (Section V-B).
3) **A five-dataset fill-rule study** with feature-level and signal-level fills, spherical-spline repair where electrode positions exist, readout refitting, and a training-side fill-bias index (Section V-E).
4) **A tuned, training-free reservoir.** The reservoir's one scale-sensitive parameter, its input gain, is chosen by nested participant-grouped cross-validation. Tuned, it matches or exceeds trained EEGNet and Riemannian classifiers on the P3 task with no trained weights in the encoder (Section V-C).
5) **A matched control for the reservoir.** A six-bin amplitude code with the reservoir's own bins separates what the spiking expansion adds from what binning alone provides (Section V-D).
6) **Reproducibility.** Code that reproduces all twelve conference table cells exactly, synthetic tests for every proposition, and aggregate results only, with no subject-level data released (Section V-A).

II. Related Work

Channel loss. Missing electrodes are handled by constant fills, by spherical-spline interpolation [6], by learned spatial filtering [2], and by training-time channel dropout [3].

Benchmarks of foundation models now report channel-dropout curves [4]. We focus on the scoring rule rather than on a new model.

Evaluation of EEG decoders. Segment-level splits leak participant identity into the test set [7], [8], cross-validation error bars are wide at typical sample sizes [9], and leakage is widespread across machine-learning-based science [10]. Unlabeled test-subject statistics, as in Euclidean alignment [11], are a legitimate adaptation when stated as such. All splits here keep participants intact.

Affective and cognitive ERPs. The late positive potential is larger for emotional than neutral pictures and is usually scored as a mean over several hundred milliseconds at parietal and occipital sites [12]. ERP CORE provides standardized paradigms and data for six components [13]. Within-participant multivariate decoding of averaged ERP subsets is an established complement to component scoring [14]. Deep networks trained within participants have decoded facial emotion from single trials [15]. We decode across participants from condition averages, a harder and different estimand.

Reservoir computing. Fixed recurrent networks map an input to a high-dimensional state read out by a simple classifier [16], [17], and their performance is often linked to the transition between ordered and chaotic dynamics [18], [19]. Here the reservoir is a reference encoder, not a proposed winner.

III. Theory

A fixed encoder maps an observation to features $x \in \mathbb{R}^D$ grouped into per-electrode blocks. The readout standardizes with training statistics, $z = D_\sigma^{-1}(x - \mu)$, and computes class logits $\ell = Wz + b$. Let S index the coordinates of the dropped electrodes and K the number of classes. Propositions 1 and 2 are from [1] and are restated for completeness.

Proposition 1 (Fill rule and re-centering)**.** *Zero fill ($x_S = 0$) and mean fill ($x_S = \mu_S$) differ by the fixed shift $z_S^{\text{zero}} - z_S^{\text{mean}} = -D_{\sigma,S}^{-1}\mu_S$. Under $x \mapsto x - c$ the clean and mean-fill inputs are unchanged, while the zero-fill input moves by $D_{\sigma,S}^{-1}c_S$. For a linear readout the zero-fill logits equal the mean-fill logits plus the constant $\delta = -W_{:,S}(\mu_S/\sigma_S)$.*

Proposition 2 (Chance floor)**.** *If $\mathbb{E}[\mathrm{BA}_{\text{pert}}] \geq 1/K$, raw retention $R = \mathrm{BA}_{\text{pert}}/\mathrm{BA}_{\text{clean}}$ satisfies $\mathbb{E}[R] \geq 1/(K\,\mathrm{BA}_{\text{clean}})$, while above-chance retention is zero at chance for every encoder.*

Proposition 3 (Zero fill to chance)**.** *Consider encoders $x_t = x + ta$ for a fixed direction a, each with its own fitted readout. Mean-fill balanced accuracy does not depend on t. If $u = -W_{:,S}(a_S/\sigma_S)$ has a unique largest coordinate k^*, zero-fill balanced accuracy equals $1/K$ for all sufficiently large t.*

Proof. Standardization removes ta, so W, b and all mean-fill logits are independent of t. The zero-fill shift is $\delta_t = \delta_0 + tu$. A test observation with mean-fill logits m is assigned to k^* once $t > \max_{k \neq k^*}(m_k + \delta_{0k} - m_{k^*} - \delta_{0k^*})/(u_{k^*} - u_k)$. A finite test set has a largest such threshold, beyond which every observation is assigned to k^*, recall is one for k^* and zero otherwise, and balanced accuracy is $1/K$. □

Proposition 4 (Per-epoch normalization of averages)**.** *Let condition c of a participant be the average of n_c trials, $a_c(t) = s_c(t) + e_c(t)$, with trial noise independent across trials and per-trial standard deviation σ at each sample. Dividing a_c by its own baseline standard deviation gives, for a long evoked-free baseline, post-stimulus values $\sqrt{n_c}\, s_c(t)/\sigma$ plus noise. The normalized contrast between two conditions is $(\sqrt{n_1}s_1(t) - \sqrt{n_2}s_2(t))/\sigma$, which vanishes when $s_1/s_2 = \sqrt{n_2/n_1}$. A single scale shared by all observations keeps the contrast proportional to $s_1 - s_2$.*

Proof. The noise of an average of n_c independent trials has standard deviation $\sigma/\sqrt{n_c}$ at each sample, whatever its spectrum, and the baseline standard deviation converges to it as the baseline lengthens. Dividing by it multiplies the signal by $\sqrt{n_c}/\sigma$. □

Fill-bias index. Proposition 1 gives a training-side predictor of how far zero fill moves a model, $B = \mathrm{spread}(\delta - \bar{\delta})/\,\mathrm{median}(\ell_{(1)} - \ell_{(2)})$, the class spread of the centered logit shift divided by the median top-two logit margin on the training data. It uses no test labels. We test whether B predicts the observed zero-fill effect in Section V-E.

IV. Materials and Methods

A. Datasets

Table I lists the five datasets. Each observation is one participant's trial-averaged ERP for one condition, the design of the conference study.

SHAPE. The Stress, Health, and the Psychophysiology of Emotion project [20] recorded 34 channels at 1024 Hz during an emotional interrupt task with neutral, pleasant and unpleasant IAPS pictures. Segments run from -200 to $+1000$ ms. An automated check found all 636 exported files to be 1229×34 with complete condition panels, and one participant with flat channels was excluded, leaving 211. The channel header was removed at export. The channel correlations match the BioSemi 32-channel layout in standard order (Mantel $r = -0.673$ against a random-order null of 0.000 ± 0.046), with the two extra channels tracking FCz and Iz. [PENDING: confirmed channel list from the lab].

TCRZEM. An independent IAPS passive-viewing cohort [21], exported as mastoid-referenced, ocular-corrected segments at 512 Hz, 32 BioSemi channels. We averaged trials by valence with arousal pooled (228 participants, median 44 trials per average). Positions are template 10–05 locations.

ERP CORE. Raw BIDS recordings [13] at 1024 Hz with digitized positions, 30 EEG channels. We used P3 (rare versus frequent letters), N400 (related versus unrelated word pairs) and N170 (faces versus cars). Data were high-passed at 0.1 Hz, re-referenced as in ERP CORE (P9/P10 mean for P3 and N400, average for N170), corrected for ocular activity by regressing bipolar vertical and horizontal EOG out of the EEG [22], epoched from -200 to $+1000$ ms, baseline-corrected, and screened with a 200 μV peak-to-peak criterion. Participants with fewer than ten trials in either condition were excluded. Without ocular correction this criterion removed 10 of 40 P3 participants, and with it 4.

TABLE I
DATASETS. K: CLASSES (SHAPE AND TCRZEM: NEGATIVE, NEUTRAL, PLEASANT). N: PARTICIPANTS. CH.: EEG CHANNELS. TRIALS: MEDIAN TRIALS PER AVERAGE. P3, N400 AND N170 ARE FROM ERP CORE. OBSERVATIONS ARE TRIAL AVERAGES PER PARTICIPANT AND CONDITION.

Dataset	K	N	Ch.	Positions	Trials
SHAPE	3	211	34	inferred	[PENDING: lab]
TCRZEM	3	228	32	template	44
P3	2	36	30	digitized	42
N400	2	35	30	digitized	58
N170	2	38	30	digitized	78.5

Each dataset reproduced its known component: TCRZEM negative minus neutral at Pz/P3/P4, 300 to 1000 ms, $+2.00$ μV ($d_z = 0.76$), N400 unrelated minus related at Cz/CPz/Pz, 300 to 500 ms, -6.54 μV ($d_z = -1.97$, negative in all 35 participants), and N170 faces minus cars at P8/PO8, 110 to 150 ms, -1.26 μV ($d_z = -0.91$).

B. Preprocessing

All datasets pass through one path: zero-phase FIR decimation to 256 Hz with stimulus onset on a kept sample, then one of four normalizations. *epoch_z* z-scores each epoch and channel over the whole epoch (the conference setting). *baseline_z* uses pre-stimulus statistics only. *pre_center* subtracts the pre-stimulus mean and divides by the whole-epoch standard deviation. *global_scale* divides every observation by one dataset-wide standard deviation. Features use post-stimulus samples only unless stated, following the recommendation to discard the baseline period.

C. Encoders

LPP is the mean amplitude from 300 to 1000 ms per channel [12]. *ERP window* is the mean amplitude in three windows per channel. *Band power* uses Welch estimates in five bands from 1 to 45 Hz. The *reservoir* is the conference encoder: 256 leaky integrate-and-fire units ($\beta = 0.05$, $\vartheta = 0.5$, spectral radius 0.9, one shared fixed weight draw), driven by each electrode trace, summarized as spike counts in six equal bins (BSC_6), and compressed by a 64-component PCA basis shared across electrodes and fitted on training participants only. BSC_6[1] sums the six bins. *Amp6*, the matched control, averages the amplitude in the reservoir's own six bins, so the reservoir's contribution beyond binning is measured directly.

Reservoir input gain. Standardization in the readout makes the linear encoders insensitive to the overall input scale, but the reservoir has a fixed firing threshold, so its spike counts depend on input amplitude. Its input is therefore multiplied by a gain $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ chosen separately in every outer training fold by a three-fold participant-grouped inner cross-validation on the training participants only, refitted on the whole outer training set, and scored on the untouched test participants. Ties go to the gain closest to 1. We report the fixed gain $\gamma = 1$ of the conference study beside the tuned result.

Trained baselines. EEGNet-8,2 [23] uses the conference recipe (Adam, learning rate 10^{-3}, 80 epochs, batch 64), with and without augmentation by random channel dropout and additive noise. Two Riemannian classifiers map ERP-prototype covariances and xDAWN covariances [24] to the tangent space at the training mean, with shrinkage estimators and the same logistic readout. All trained models see the same post-stimulus window, participants and removed electrodes as the fixed encoders.

D. Channel loss

In each fold a fixed fraction of electrodes (10, 30 or 50%) is removed in the test participants, with the same draws for every encoder. Feature-level fills replace the dropped blocks by zero, the training mean, a k-nearest-neighbor estimate ($k = 10$), the mean of retained channels, or a spherical-spline estimate [6]. Signal-level removal zeroes the dropped traces before encoding, and signal-level spline repair interpolates them before encoding. Refit trains the readout on the retained channels only.

E. Validation

Five repeats of five-fold participant-grouped stratified cross-validation, train-only standardization, and a balanced ℓ_2 logistic readout ($C = 1$). Out-of-fold probabilities are pooled across repeats before scoring, and 95% intervals come from 4000 resamples of participants, each keeping all its observations. Encoder and fill contrasts are paired under the same resamples. Within-participant label permutation (1000 permutations, headline estimator) gives the null distribution. [PENDING: permutation run].

V. RESULTS

A. Exact reproduction of the conference results

Rebuilt from the raw SHAPE files with the conference preprocessing, the new code reproduced all twelve conference table cells to six decimals (clean, 30% zero, mean, kNN and spatial fills, and signal-level removal, for band power, ERP windows and the reservoir). The match also confirms that the cross-validation splits are identical.

B. Normalization artifacts

Offset leak into a pre-stimulus window. The conference reservoir window spanned -160 to $+74$ ms around onset. Under whole-epoch z-scoring its clean balanced accuracy was 0.487. Under baseline-only scaling, which keeps post-stimulus samples from influencing pre-stimulus values, it fell to 0.305 [0.272, 0.340], against a chance level of 0.333 (0.340 [0.303, 0.376] with FIR decimation), and the matched Amp6 code fell from 0.536 to 0.308. With post-stimulus windows the reservoir kept most of its accuracy under baseline-only scaling (late window 0.602 to 0.566). The conference reservoir accuracy therefore came almost entirely from post-stimulus signal reaching the window through the normalization, a caveat the conference paper raised but could not quantify. Anti-aliased decimation changed no conclusion (every cell within five points of its legacy counterpart), consistent with the 30 Hz low-pass already applied upstream.

Trial counts. In the P3 task the rare average contains about four times fewer trials than the frequent one. In microvolts, rare minus frequent at Pz from 300 to 1000 ms gave $d_z = 1.05$. After baseline z-scoring it gave $d_z = 0.05$, and LPP decoding fell from 0.708 to 0.500 [0.389, 0.611], as Proposition 4 predicts. Under *global_scale* the LPP score decoded at 0.708. [PENDING: effect on SHAPE, which needs per-condition trial counts].

Scale normalization itself helps cross-participant decoding. On SHAPE with FIR decimation, ERP-window features scored 0.621 under whole-epoch z-scoring and 0.567 under *pre_center*, which keeps the scale normalization but removes the offset leak, so about five points of that score are attributable to the leak. Baseline-only scaling lowered every encoder, including post-stimulus windows, so whole-epoch z-scoring also helps through a per-epoch amplitude normalization. [PENDING: full normalization grid with paired CIs].

C. Gain-tuned reservoir against trained models

Table II compares clean accuracy on the three ERP CORE tasks. On P3, nested gain selection chose $\gamma = 2$ in 17 of 25 outer folds, $\gamma = 4$ in 7 and $\gamma = 1$ in one, so the conference gain was too low for this input scale. The tuned reservoir reached 0.889, against 0.861 at $\gamma = 1$ (paired $+0.028$ [-0.042, $+0.111$]). It exceeded the LPP score by $+0.181$ [$+0.083$, $+0.278$] and the matched Amp6 code by $+0.042$ [-0.028, $+0.111$], and it was the most accurate encoder on the task in point estimate, ahead of the best Riemannian classifier (0.875) and EEGNet (0.847) without any trained weights in the encoder. On N400 tuning made no difference (0.829 at both settings, paired $+0.000$ [-0.057, $+0.057$]) and the chosen gain varied between folds (2 in 10, 0.5 in 9, 1 in 4, 4 in 2), so this task is insensitive to the input scale. There the tuned reservoir still exceeded the LPP score ($+0.129$ [$+0.029$, $+0.229$]) but trailed EEGNet and the ERP-covariance classifier by 7 to 9 points. On N170 the inner search chose $\gamma = 1$ in 15 of 25 folds and tuning did not help (0.671 against 0.684, paired -0.013 [-0.066, $+0.039$]). Tuning therefore paid off only on P3, the task dominated by a large slow wave. The comparisons with trained models are point estimates from the same splits. [PENDING: paired tuned-reservoir versus trained-model contrasts, and gain-tuned results on TCRZEM and SHAPE].

D. What the reservoir adds beyond binning

On SHAPE we crossed two decimations, two normalizations and four windows (16 cells). Two results favor the reservoir. First, its six-bin temporal code carried information beyond the total spike count in every post-stimulus cell (BSC_6 minus BSC_6[1] $+0.035$ to $+0.120$, interval above zero in 11 of 12 cells), whereas the two were identical in the conference window. Second, once the offset leak was removed (baseline-only scaling), BSC_6 exceeded the ERP-window features in the late and full post-stimulus windows by $+0.051$ to $+0.082$ (all four intervals above zero, for example $+0.082$ [$+0.039$, $+0.128$] with FIR decimation and the full window). The matched control was nonetheless stronger: Amp6 exceeded

TABLE II
CLEAN BALANCED ACCURACY ON ERP CORE (CHANCE 0.5), *global_scale*, POST-STIMULUS WINDOW, SAME PARTICIPANT-GROUPED SPLITS FOR EVERY ROW. BSC_6 ROWS USE NO TRAINED WEIGHTS IN THE ENCODER. TS: RIEMANNIAN TANGENT SPACE. BOLD: BEST PER TASK.

Encoder	P3	N400	N170
BSC_6, tuned γ	**0.889**	0.829	0.671
BSC_6, $\gamma = 1$	0.861	0.829	0.684
Amp6	0.847	0.771	0.618
LPP 300–1000 ms	0.708	0.700	0.553
EEGNet	0.847	0.900	0.776
EEGNet + aug.	0.847	0.886	0.763
TS, ERP cov.	0.875	**0.914**	**0.816**
TS, xDAWN	0.861	0.829	0.763

BSC_6 in 9 of 12 post-stimulus cells by 4.6 to 8.8 points, with the early window under baseline-only scaling a tie ($+0.006$ [-0.039, $+0.054$]). On ERP CORE P3 under *global_scale*, BSC_6 minus Amp6 was $+0.014$ [-0.083, $+0.111$], and BSC_6 exceeded the LPP score by $+0.153$ [$+0.042$, $+0.264$]. On N400 under *global_scale*, BSC_6 minus Amp6 was $+0.057$ [-0.043, $+0.143$], while BSC_6 exceeded the LPP score ($+0.129$ [$+0.029$, $+0.229$]), ERP windows ($+0.114$ [$+0.014$, $+0.214$]) and band power ($+0.171$ [$+0.071$, $+0.271$]). On N170 under *global_scale*, BSC_6 minus Amp6 was $+0.066$ [-0.039, $+0.184$] and BSC_6 exceeded ERP windows ($+0.171$ [$+0.053$, $+0.289$]). Across the three ERP CORE tasks BSC_6 led Amp6 by 1 to 7 points in point estimate, but no paired interval excluded zero. On TCRZEM, the larger affective cohort, the matched control was better: BSC_6 minus Amp6 was -0.045 [-0.086, -0.004] under *global_scale* (-0.039 [-0.085, $+0.003$] under baseline_z), and BSC_6 was indistinguishable from the LPP score (-0.012 [-0.050, $+0.028$]). For affective picture viewing, the spiking expansion therefore loses information that binning alone keeps. Whether it adds information for brief or shape-defined components such as the N400 and N170 remains open at the ERP CORE sample sizes. [PENDING: paired contrasts for SHAPE]. With 35 to 38 participants the paired intervals on ERP CORE are about ± 10 points wide, so only the two large cohorts can resolve differences of a few points.

Weight draws. At spectral radius 0.9 three independent weight draws gave 0.585, 0.586 and 0.585 on SHAPE (FIR decimation, baseline_z, post-stimulus window), so the reservoir accuracy does not hinge on one draw. The dependence on spectral radius differed between draws, so we do not report an optimal operating point. [PENDING: all ten draws, threshold density, branching threshold, p1b].

E. Fill rules across datasets

The reservoir embedding was near-centered on all three ERP CORE tasks, with mean $|\mu_i|/\sigma_i$ of 0.14 to 0.17 against 0.43 to 1.03 for the other encoders and a fill-bias index of 0.01 to 0.03 against 0.23 to 1.13. Its mean-minus-zero fill shift at 30% dropout was $+0.013$ to $+0.014$. The other encoders shifted more and in either direction, as Proposition 1 allows: on N400 mean fill was worse than zero fill for ERP windows

(-0.057 [-0.114, -0.014]). [PENDING: fill-bias index versus observed shift across all datasets and rates, Proposition 3 re-centering experiment, spline and refit summaries].

VI. Discussion

Reporting checklist. We recommend that channel-loss studies state:

1) the fill rule, with at least one in-distribution fill and, where positions exist, spline repair,
2) the normalization, and whether its statistics use post-stimulus samples,
3) for averaged ERPs, the trial counts per condition and whether any per-epoch scaling depends on them,
4) absolute accuracy with participant-level intervals beside raw and above-chance retention,
5) the augmentation state of trained models.

Relation to the conference paper. New here: Propositions 3 and 4, the fill-bias index, four additional datasets, the matched Amp6 control, the normalization analysis, spherical-spline and refit conditions, multiple reservoir weight draws, nested selection of the reservoir input gain, trained EEGNet and Riemannian baselines, and exact reproduction code. The conference results are reproduced, and the conference reservoir accuracy is reinterpreted as a normalization effect.

Limitations. Observations are condition averages, so we do not estimate single-trial performance or within-participant decoding [14]. The ERP CORE samples are small. SHAPE positions are inferred until the lab confirms the channel list, and TCRZEM positions are templates. The reservoir uses one architecture. The gain search was introduced in this study after the fixed-gain results were known. Because the gain is chosen inside each training fold, test participants never inform it, but the decision to tune it was not prespecified, and we therefore report the fixed-gain results beside it. EEGNet uses one recipe without a hyperparameter search, so its accuracies may be conservative. Clinical labels play no role, and nothing here is a diagnostic claim.

VII. Conclusion

[PENDING: write after Phase 1 and 2 results are final].

Data Availability

SHAPE is available from the Laboratory for Clinical Affective Neuroscience, Stony Brook University [20]. TCRZEM [21] and ERP CORE [13] are public. Code and aggregate results: [PENDING: repository DOI at release].

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